OVERTURN.

The denial, answered with evidence.

- Zetaris · the data path
- Meterless · H-MEM + World Model
- Meterless Nightshift · primary AI
- NVIDIA Nemotron · Critic + Advocate

CASE CLM-2026-0915

Ava.

34 · Sydney

Northbridge Health

MRI-LS-001 · M54.5

84 /100

SOLID · CONDITIONAL

THE PROBLEM

A denial is not a decision. It is paperwork.

Ava is 34. Her MRI was denied with four reasons. To appeal, she has to read the policy, find the clause that actually applies, reconcile the procedure and diagnosis codes, and answer each reason with a citation — inside a 60-day window.

4
reasons in one denial letter

60
days from denial to appeal deadline

1
required document genuinely missing

WHO IT IS FOR

01 **Patients and families**
People holding the letter, who are not expected to know what POL-4.3 means.

02 **Patient advocates and claims navigators**
The people who do this for someone else, case after case.

03 **Small clinics and advocacy nonprofits**
Teams with no in-house appeals department and no spare lawyer.

ONE DENIAL · FOUR REASONS

Ava.

34 · Sydney

INSURER	Northbridge Health
CLAIM	CLM-2026-0915
SERVICE	MRI lumbar spine, no contrast MRI-LS-001
DIAGNOSIS	M54.5 · low back pain
DENIED	2026-10-01
DEADLINE	2026-11-30

DEN-01	The service is experimental or investigational. Cites POL-4.3 — the exclusion that applies to spinal fusion.	Rebutted
DEN-02	MRI-LS-002, with contrast, is not covered without prior authorisation. The file says MRI-LS-001.	Rebutted
DEN-03	No conservative treatment is documented. The record has 12 physiotherapy sessions over 12 weeks.	Rebutted
DEN-04	The specialist referral letter was not provided. It is genuinely absent from the file.	Missing

Three reasons are contradicted by the file. The fourth is missing — and stays missing. **It is never fabricated.**

WHAT OVERTURN DOES

Read the file. Find the contradiction. Answer it.

OVERTURN is an appeal agent, not a chat assistant. It reads the denial, the policy, the clinical record, the EOB, the code tables and the document checklist — then drafts an internal appeal packet a human can send.

01

Read

Nine sources, queried through one read-only tool. Policy clauses, denial reasons, records, EOB, procedure codes, document checklist, ICD-10.

`run_sql`

02

Reconcile

Every claim becomes an evidence line with a source, a row and a hash. 47 lines for Ava. The Reconciler is the only writer.

`evidence ledger`

03

Re-query

When the file disagrees with itself, the Critic issues a precise query and the Reconciler runs it. Bounded, traced, five times.

`max 4 + max 3`

04

Report

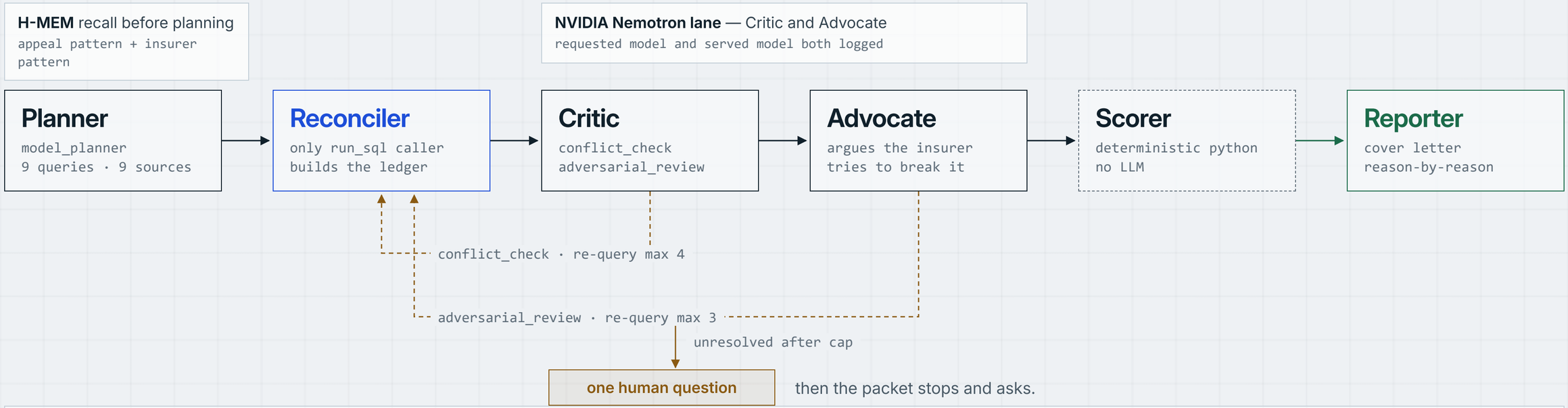
A cover letter, a reason-by-reason rebuttal, a next-best-action, and a receipt that shows exactly which line supports which sentence.

`packet + receipt`

OUTPUT `appeal_packet.md` `appeal_packet.json` `receipt.html` `trace.jsonl`

Five agents. Two loops. One shared case.

The Reconciler is the only agent that calls `run_sql`. The Critic and Advocate can only ask for a re-query. Every hop is written to `trace.jsonl` before the next step.



NO UNCITED SENTENCES

Every sentence points at a line.

47	evidence lines, each with a source table and row id
5	re-queries executed by the Reconciler
1	SHA-256 chain from genesis to head

SHA256-CHAIN-V1 · CANONICAL JSON · UTF-8

GENESIS	00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000	
L0001	6f6d2325a3dc…	policy_clauses:POL-10.1
L0002	5b54f483faf1…	policy_clauses:POL-12.4
L0006	325038afcf44…	policy_clauses:POL-4.3
HEAD · 47	387f5f91b90a…	the packet's head hash

Recompute the chain from the emitted lines. If one byte moves, the head moves.

Two citations minimum
A rebuttal without two sources is unsupported, and the score says so.

Counterfactual by source
Remove a table, recompute the score. The packet reports the delta.

The letter stays missing
DOC-003 is listed, not invented. The appeal says what to send.

APPEAL STRENGTH

84 / 100

Solid.

Conditional — one required document missing.

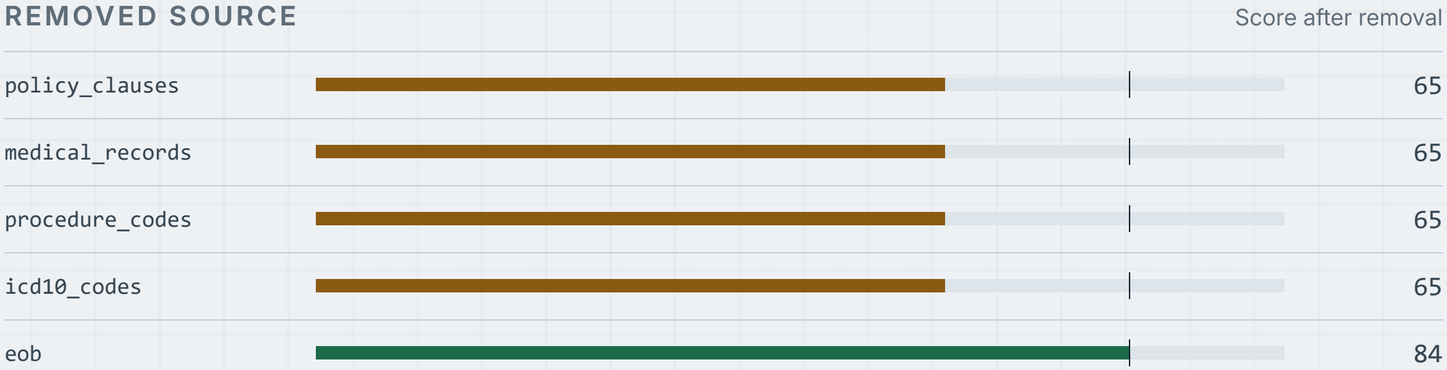
```
100 start
- 15 one missing document
+ 5 all non-missing reasons cited twice
= 90 raw
capped at 84 because DOC-003 is missing
```

NEXT BEST ACTION

Obtain the specialist referral letter (DOC-003). Supplying it raises the file to **`100 / ready`**. The packet will not fabricate it.

THE DEPENDENCY TEST

Remove one source at a time and recompute. If the score does not move, the source was corroboration — not proof.



The marker is the actual 84. Removing the EOB leaves the code and record argument intact, so the score does not move. That is honest, not a bug.

- 3 of 4 reasons rebutted
- 1 missing · 0 unresolved

THE STACK

A thin orchestrator. Nothing magical in the middle.

Five agents, hard caps and a JSONL trace. The loop is custom code, not a framework — so every decision is inspectable and every model call is a single adapter.

LANGUAGE	`Python 3.11+`	stdlib-first; httpx and python-dotenv are the only runtime dependencies.
ORCHESTRATION	`thin custom loop`	Planner, Reconciler, Critic, Advocate, Reporter — with hard iteration caps.
LIVE DATA	`Zetaris REST`	run_sql is the only data tool. Read-only validation before any query leaves the process.
OFFLINE DATA	`SQLite mirror`	SAMPLE_MODE runs the same SQL with no credentials, in about one second.
AI PROVIDER	`Meterless Nightshift`	OpenAI-compatible contract; model discovery and served-model logging.
ADVERSARY	`NVIDIA Nemotron lane`	Critic and Advocate pinned to the Nemotron model when configured.
STATE & MEMORY	`World Model + H-MEM`	One shared case inside a run; appeal and insurer patterns across runs.
OUTPUT & TESTS	`MD · JSON · HTML · JSONL`	A signable packet, an evidence receipt, a full trace. 36 passing tests.

THE ONLY DATA PATH

Seven sources. One semantic layer.

policy_clauses	clauses	denial_reasons	reasons
medical_records	records	eob	explanation
procedure_codes	codes	claim_documents	checklist
icd10_codes	descriptions	usl_claim_review	USL
appeal_evidence	view	claim_facts	measures
dq_claim_integrity	5 DQ rules		

Unified Semantic Layer
usl_claim_review, appeal_evidence and claim_facts, with its DDL in the registration script.

Five quality rules
dq_claim_integrity checks orphan clauses, unknown codes, undated records, fabricated documents and missing-count drift.

One caller
Only the Reconciler invokes run_sql. No other agent holds the tool.

THE LIVE CONTRACT

`jdbc:zetaris:lightning@…:10000`

The live run used the JDBC endpoint: 13 read-only calls, USL + five DQ rules pass. The REST proxy contract is also implemented and tested.

```
# a real query from out/trace.jsonl
SELECT * FROM denial_reasons
ORDER BY reason_id
-- 4 rows · 3 rebutted, 1 missing
```

WHY IT IS LOAD-BEARING

Policy, denial, record, EOB, codes, checklist and ICD-10 descriptions all arrive through the same tool. There is no second data path to drift.

The packet's provenance is therefore the query's provenance.

METERLESS, THREE WAYS

The provider, the case, and the memory.

Nightshift is the primary AI provider. H-MEM and the World Model are what make the second appeal smarter than the first.

METERLESS NIGHTSHIFT

Primary AI provider.

The model lane for every agent call.

- OpenAI-compatible chat contract
- `meterless-app` and `meterless-scope` headers on every request
- `/v1/models` discovery before selection
- Requested and served model both written to the trace
- Critic and Advocate pinned to the Nemotron model

METERLESS WORLD MODEL

The shared case.

One object, not a message bus.

- Holds evidence lines, denial reasons, missing documents and the packet
- The Reconciler writes; the Critic and Advocate read
- Re-query mutates the same case the next agent sees
- Adapter tested against the Meterless HTTP contract

METERLESS H-MEM

Memory across runs.

What this insurer keeps doing.

- Recalls the claimant's appeal pattern
- Recalls the insurer's denial pattern
- 8 prior memory items recalled in the sample run
- Two recall kinds: appeal pattern, insurer pattern

SCOUT INTENT A deterministic router selects `denial_appeal`, `billing_error` or `prior_auth` before any agent runs.

THE COMBINATION

The sponsors are not logos. They are the argument.

01 **Remove the World Model.**

The Critic has no shared case to read. Re-query stops being stateful, and the agents start passing summaries instead of evidence.

02 **Remove H-MEM.**

Every appeal starts cold. The insurer's denial pattern is relearned on every run, and the second case costs the same as the first.

03 **Remove Nightshift's Nemotron lane.**

The adversary and the rebuttal collapse into one voice. Nobody is left to try to break the appeal before the insurer does.

Two models argue. One deterministic referee scores. **The human signs.**

WHAT STAYS OUTSIDE THE MODEL

01 The appeal-strength score — deterministic Python, no LLM.

02 The SQL guard — writes, DDL and multiple statements rejected.

03 The SHA-256 chain — recomputable from the emitted evidence.

04 The missing letter — listed as missing, never generated.

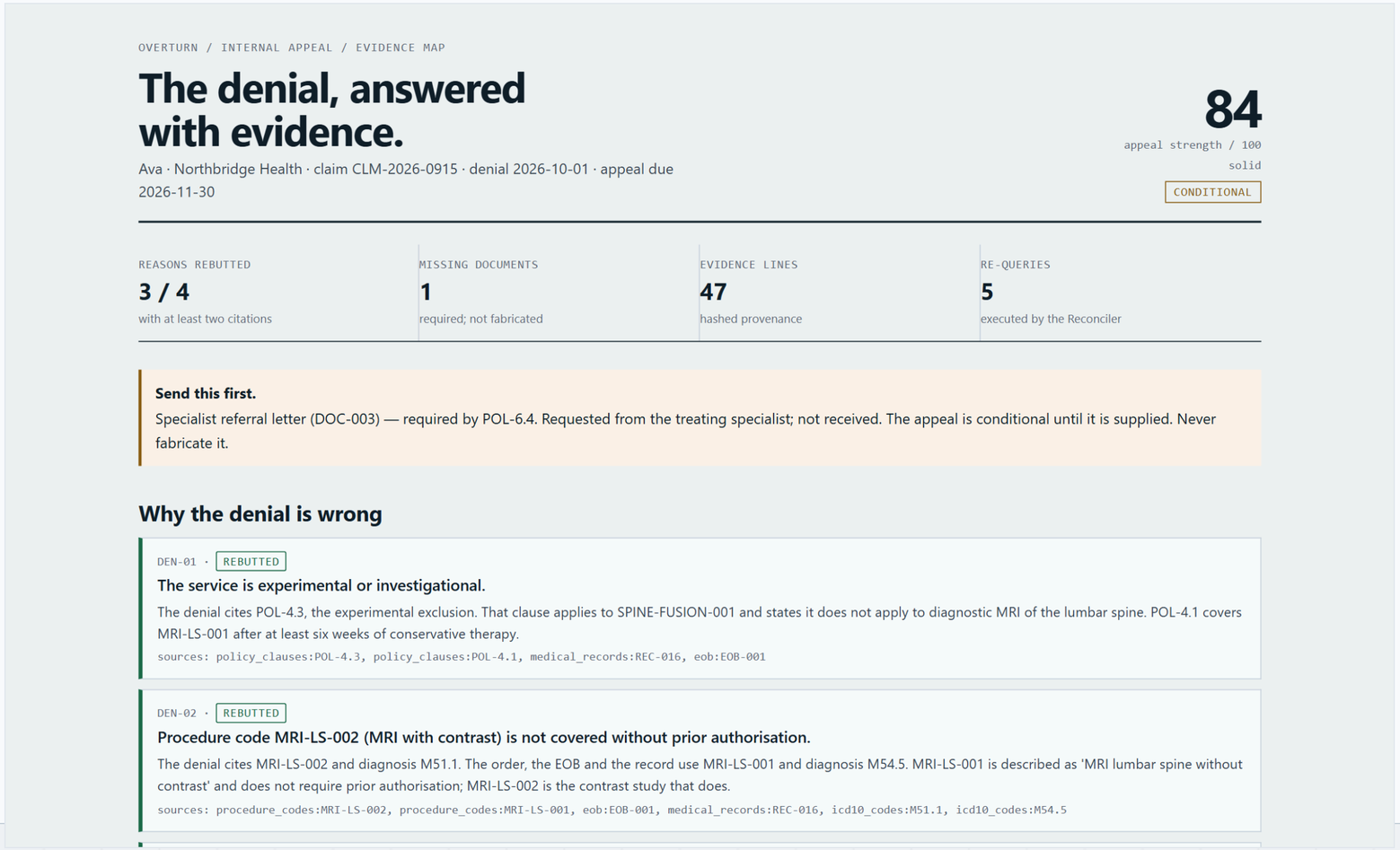
THE PRODUCT

The packet a human signs.

Every claim in the packet links back to a line in the receipt. The receipt is generated from the same run — not written by hand.

out/receipt.html

evidence map · 84 / conditional



OVERTURN / INTERNAL APPEAL / EVIDENCE MAP

The denial, answered with evidence.

Ava · Northbridge Health · claim CLM-2026-0915 · denial 2026-10-01 · appeal due 2026-11-30

84

appeal strength / 100

solid

CONDITIONAL

REASONS REBUTTED	MISSING DOCUMENTS	EVIDENCE LINES	RE-QUERIES
3 / 4	1	47	5
with at least two citations	required; not fabricated	hashed provenance	executed by the Reconciler

Send this first.

Specialist referral letter (DOC-003) — required by POL-6.4. Requested from the treating specialist; not received. The appeal is conditional until it is supplied. Never fabricate it.

Why the denial is wrong

DEN-01 · REBUTTED

The service is experimental or investigational.

The denial cites POL-4.3, the experimental exclusion. That clause applies to SPINE-FUSION-001 and states it does not apply to diagnostic MRI of the lumbar spine. POL-4.1 covers MRI-LS-001 after at least six weeks of conservative therapy.

sources: policy_clauses:POL-4.3, policy_clauses:POL-4.1, medical_records:REC-016, eob:EOB-001

DEN-02 · REBUTTED

Procedure code MRI-LS-002 (MRI with contrast) is not covered without prior authorisation.

The denial cites MRI-LS-002 and diagnosis M51.1. The order, the EOB and the record use MRI-LS-001 and diagnosis M54.5. MRI-LS-001 is described as 'MRI lumbar spine without contrast' and does not require prior authorisation; MRI-LS-002 is the contrast study that does.

sources: procedure_codes:MRI-LS-002, procedure_codes:MRI-LS-001, eob:EOB-001, medical_records:REC-016, icd10_codes:M51.1, icd10_codes:M54.5

01

84 / 100 · conditional

The verdict sits at the top, next to the case and the appeal deadline.

02

3 of 4 · 47 lines · 5 re-queries

The counts come from the run, not from a template.

03

“Send this first.”

The missing letter is named, with its policy clause and request date.

04

Four reasons, four receipts

Each rebuttal carries its sources; the missing one is marked, not argued.

`appeal_packet.md` `appeal_packet.json` `trace.jsonl`

TEST SUITE

36

tests pass in about four seconds, with no credentials and no network.

- 37 passed

CONTRACT TESTS — NO LIVE KEYS NEEDED

Zetaris path, bearer auth, org header, UUID request id, `{"sql","limit"}` body and row extraction. Nightshift `meterless-app` / `meterless-scope` headers, model discovery, selected Nemotron model and served model in the trace.

FEATURE TESTS

Both loop caps, the four planted conflicts, the World Model and H-MEM adapters, the deterministic score, the counterfactual rules and the SHA-256 chain.

OFFLINE END-TO-END

`python -m overturn.run --case ava --out out/` runs the whole pipeline against the SQLite mirror and produces the packet above.

SYNTHETIC BY DESIGN

No real patient, insurer, provider, claim or PHI. The denial, policy, records and codes are generated deterministically, so every run is inspectable.

`test_zetaris_client.py`	live SQL contract	`test_nightshift_client.py`	headers · discovery	`test_meterless_adapters.py`	World Model + H-MEM
`test_control_loop.py`	both bounded loops	`test_scorer.py`	84 → 100 · buckets	`test_provenance.py`	chain recompute
`test_conflicts.py`	four planted conflicts	`test_router.py`	intent routing	`test_data_integrity.py`	source fixtures

Live run: Zetaris JDBC (13 read-only queries, USL + five DQ rules) and Nightshift (11 agent calls, served by nvidia/nemotron-3-ultra-550b-a55b). H-MEM/World Model used local mirrors.

WHAT'S NEXT

From one appeal to an appeals desk.

The same loop generalises: read a file, find the contradiction, answer it with evidence. The next work is intake, scale and outcome feedback.

01	**Run the live path**	Zetaris platform bundle and Nightshift credentials; prove the contracts against real tables.
02	**Intake that reads paper**	OCR and layout parsing for scanned denial letters, faxed records and EOBs.
03	**A policy graph per insurer**	Clause versions, effective dates and which exclusions actually apply to which service.
04	**An advocate workspace**	Queue, review, sign and send — with a per-case audit export built from the trace.
05	**Outcome feedback**	Log the insurer's answer in H-MEM and learn which arguments move which payer.

OPEN AGENT HACKATHON 2026

OVERTURN.

The denial, answered with evidence.

github.com/Aryamanironman/overturn

Zetaris | Meterless · Nightshift · H-MEM · World Model | NVIDIA Nemotron